Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	03 January 2026
Team ID	LTVIP2026TMIDS65757
Project Name	TransLingua – AI-Powered Multi-Language Translator
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Application Setup	USN-1	As a developer, I can set up the Python virtual environment and install dependencies.	2	High	Team
Sprint-1	UI Development	USN-2	As a user, I can enter text for translation.	2	High	Team
Sprint-1	Language Selection	USN-3	As a user, I can select source and target languages.	2	High	Team
Sprint-1	API Integration	USN-4	As a system, I can integrate Google Gemini API for translation.	5	High	Team
Sprint-2	Security Implementation	USN-5	As a developer, I can store API keys securely using environment variables.	3	High	Team
Sprint-2	Output Display	USN-6	As a user, I can view translated text instantly.	2	High	Team
Sprint-2	Testing	USN-7	As a developer, I can perform functional testing.	3	Medium	Team
Sprint-2	Documentation	USN-8	As a developer, I can prepare GitHub repository and documentation.	5	Medium	Team

Total Story Points per Sprint

Sprint-1 = 2 + 2 + 2 + 5 = **11 Story Points**
Sprint-2 = 3 + 2 + 3 + 5 = **13 Story Points**

Total Story Points = 11 + 13 = **24**

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	**Total Story Points**	**Duration**	**Sprint Start Date**	**Sprint End Date (Planned)**	**Story Points Completed**	**Sprint Release Date**
Sprint-1	11	10 Days	5 Jan 2026	14 Jan 2026	11	14 January 2026
Sprint-2	13	10 Days	15 Jan 2026	24 Jan 2026	13	24 January 2026
Sprint	**Total Story Points**	**Duration**	**Sprint Start Date**	**Sprint End Date (Planned)**	**Story Points Completed**	**Sprint Release Date**
Sprint-1	11	10 Days	5 Jan 2026	14 Jan 2026	11	14 January 2026
Sprint-2	13	10 Days	15 Jan 2026	24 Jan 2026	13	24 January 2026
Sprint	**Total Story Points**	**Duration**	**Sprint Start Date**	**Sprint End Date (Planned)**	**Story Points Completed**	**Sprint Release Date**
Sprint-1	11	10 Days	5 Jan 2026	14 Jan 2026	11	14 January 2026
Sprint-2	13	10 Days	15 Jan 2026	24 Jan 2026	13	24 January 2026

4. Velocity Calculation

Velocity = Total Story Points Completed / Number of Sprints

Total Story Points = 24
Number of Sprints = 2

Velocity = 24 / 2 = 12 Story Points per Sprint

If sprint duration is 10 days,

Average Velocity per Day = 12 / 10 = 1.2 Story Points per Day

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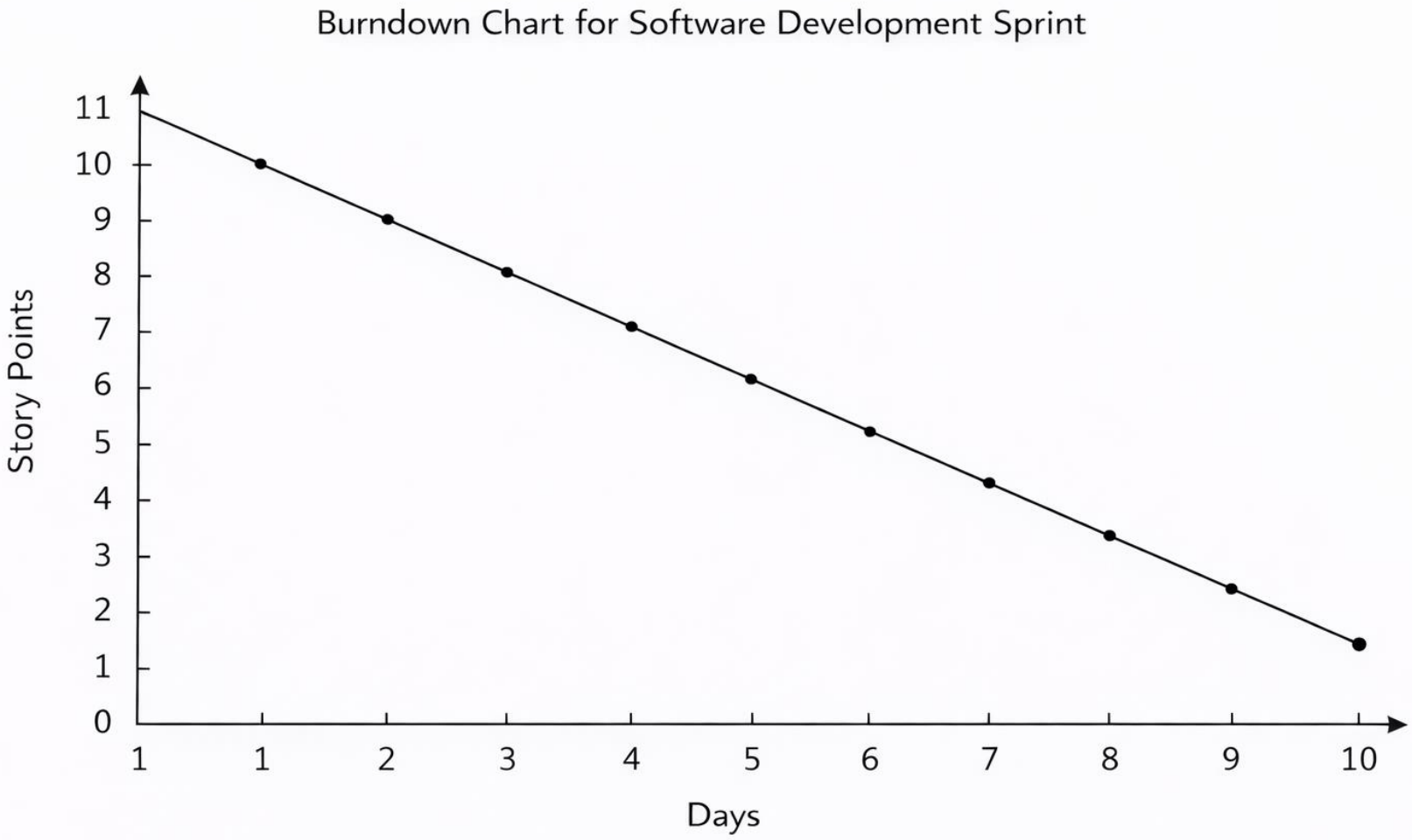
5. Burndown Chart Description

A burndown chart represents the remaining work against time during a sprint.

For this project, each sprint started with its respective total story points and gradually reduced to zero by the planned sprint end date. This indicates systematic progress and effective sprint management.


Burndown Chart for Software Development Sprint
Story Points
11
10
9
8
7
6
5
4
3
2
1
0
1
1
2
3
4
5
6
7
8
9
10
Days

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts